

AI-POWERED TRAIN SAFETY SYSTEM

25-26J-134

Project Proposal Report

Prarthana L.D.R – IT22291578

B.Sc. (Hons) Degree in Information Technology

Specializing in Data Science

Faculty of Computing

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

AI-POWERED TRAIN SAFETY SYSTEM

25-26J-134

Project Proposal Report

Prarthana L.D.R – IT22291578

B.Sc. (Hons) Degree in Information Technology

Specializing in Data Science

Faculty of Computing

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

		Date
Signature		
Signature of the Supervisor		
Signature of the Co-Supervisor		

ABSTRACT

Sri Lanka's railroad transportation network is marred by serious safety problems at unmanned level crossing points and low-visibility track sections where footpeople, animals, primarily elephants, regularly obstruct train paths. Such incidents frequently culminate in fatal crashes, infrastructures destruction, and disruption of service. Aims at conceiving a real-time object detection and crash prevention system for railways based on a fusion between computer-vision and multimodal sensors is this proposed research. Video datasets in addition to image datasets are used for training a deep model for object detection while humans, elephants, and vehicle classification. TensorFlow Lite / Pytorch Mobile based-lightweight deployment pipeline shall enable real-time onboard processing on smartphones mounted on locomotive vehicles. To enable enhanced detection accuracy, besides object detection using HC-SR04 ultrasonic sensor for object detection and distance approximation and AMG8833 infrared heat sensing arrays for detection of warm-bodied obstacles in low-visibility scenarios such as fog at night based on object heat signature acquisition as a frame for determination of object movement; a scaled model based on a 4- Wheel Smart Car Chassis kit shall mimic a railroad vehicle with encoded speed encoders, motor controller driver units, and control logic for a collision-free track. Computations for obstacle distance, train speed, and braking needed for safe distances for blocking for determination for warning alarms issue for drivers or for control centers. Further still, it computes normal arrival times versus delayed arrival times between stations based on object detection occurrences. The constructed method holds novelty in a fusion between heat sensing infrared arrays, ultrasonic ranging, and computer-vision based deep learning models in a railroad safety application in providing a low-cost yet scalable safety intervention for mitigating accident occurrence on Sri Lanka Railway's network.

In addition, it involves integration of train speed and braking distance estimation in calculating obstacles for real-time risk estimation and variable stop-go decisions. It is a first in the Sri Lankan railroad environment in bringing together multi-sensor fusion with predictive braking kinematics, which enhances further active accident avoidance as well as safe operations

Table of Contents

DECLARATION	3
ABSTRACT	4
INTRODUCTION	6
BACKGROUND AND LITERATURE SURVEY	8
BACKGROUND	8
LITERATURE SURVEY	8
RESEARCH GAP	10
RESEARCH PROBLEM	11
OBJECTIVES	12
MAIN OBJECTIVE	12
SPECIFIC OBJECTIVES	12
METHODOLOGY	14
OVERALL SYSTEM ARCHITECTURE	14
OBJECT DETECTION SYSTEM	15
FUNCTIONAL REQUIREMENTS	19
NON-FUNCTIONAL REQUIREMENTS	20
EXPECTED TEST CASES	20
REFERENCES	23

Figure 1: System Architecture Diagram.....	14
Figure 2: Object detection system Component - High Level System Architecture Diagram.....	15
Figure 3: Train movement with sensor wave	16
Figure 4:Distance calculation formula	17

INTRODUCTION

Railway transport remains one of the most vital modes of mobility in Sri Lanka, supporting both The modes of Railway transportation remain one of Sri Lanka's most essential modes of movement, allowing for passenger as well as cargo movement within the nation. However, safety within the system remains significantly undermined by recurrent accidents at unmanned level crossing points as well as sections where visual exposure is diminished. According to national Railway statistics, pedestrian-vehicle-wild animal in particular elephant-type accidents are a frequent occurrence, with ensuing loss of life, disruption in service, and significant economic losses. Automated surveillance is not employed, further obscuring risks since in a majority of situations, train operators are left with minimal time for response when confronted with an obstruction unexpectedly, mostly at night or in unsuitable weather conditions.

Traditional safety schemes for railways are based on mechanically or human-monitored watching or obstacles-barriers, which are costly if imposed in large scales and inapplicable in rural settings. Upgraded research in sensor technology as well as in computer visions creates a possibility for constructing more adaptable safety mechanisms which are smarter. Object detection models such as YOLO prove high capability in real-time classification of obstacles in environments which are cluttered. Based on visual information alone might prove insufficient in low-light or foggy environments. To compensate for this disadvantage, a multimodal sensing approach which uses a fusion of ultrasonic sensor module up to distance measurement with thermal imaging proves effective in maximizing accuracy and robustness in obstacle detection.

The research recommends developing an integrated safety system for railways based on three major technologies:

- Computer Vision Models were trained on image datasets (containing elephants, humans, vehicles, etc.) in order to recognize and classify real-time obstacles.
- Ultrasonic Distance Sensing (HC-SR04) is used for calculating relative close proximity of objects found towards a moving train for distance computation.
- Thermal Imaging (AMG8833) for detection of warm-bodied objects so you can detect at night or in low-visibility foggy settings in addition to object motion.

A scaled-down prototype shall be fitted in a 4-Wheel Smart Car Chassis test bed for an imitation of train behavior including speed encoders and motor drivers for stop control mechanisms. Safe braking distances shall be calculated by the system, warning messages shall be generated for drivers, and arrival time delays at stations shall be calculated upon object detection. The study is original in its combination of compact deep learning models with multimodal sensors for use in developing nations for safety in railways. Compared with conventional barrier or single-sensor approaches, our

proposed technique offers a low cost and scalable safety system for use in Sri Lanka's unique railway environment. Its outcome is expected to lay a groundwork for real-world application with a minimum number of collisions, loss of life prevented, and maximum effectiveness in operations.

BACKGROUND AND LITERATURE SURVEY

Background

Sri Lanka's railway network extends across busy commuter routes and long rural sections, where unmanned level crossings and low-visibility track segments (caused by night, fog, sharp curves, cuttings, and overgrown track edges) are common. These conditions often lead to collisions with pedestrians, vehicles, and wild elephants, resulting in fatalities, delays, and damage to trains.

Conventional safety measures such as barriers, manned gates, and camera surveillance from control centers are expensive to scale and difficult to maintain in rural areas. This highlights the need for self-contained, train-mounted perception and decision-support systems that:

- Work reliably under variable lighting and weather,
- Do not depend on wayside power or connectivity,
- Deliver real-time hazard detection directly to the driver.

Therefore, an integrated multi-sensor obstacle detection system emerges as a practical and scalable solution to improve railway safety in Sri Lanka.

Literature Survey

Vision Based Obstacle Detection on Rail

- Modern hazard detection is dominated by deep learning approaches:

YOLO: fast, single-shot detectors, widely used for trespassing and vehicle detection, well suited for real-time edge devices.

- Challenges: camera-only systems struggle in low light, fog, and with headlight glare.

Thermal Sensing to Address Poor Visibility

- AMG8833 8×8 thermal array detects heat-emitting objects up to 7 m.
- Advantages: functions in night/fog, compact, and low power.
- Despite coarse resolution, interpolation and temporal filtering (Gaussian smoothing, exponential moving average) stabilize heatmaps.
- Prior studies show even low-resolution grids can support binary presence detection and rough classification (human/animal/no object) using lightweight CNNs or SVMs.

Ultrasonic Ranging on Moving Platforms

- HC-SR04 ultrasonic sensor: range 2–400 cm, accuracy 3 mm.
- Limitation: moving trains cause distance bias since the train shifts during echo return.
- Mitigation:
 - Use for short-range gating (<3–4 m) where errors are minimal.
 - Apply multi-ping averaging with motion compensation (odometry).
 - Use ultrasonic as a near-range confirmation layer rather than the sole detector

Sensor Fusion for Reliable Hazard Assessment

- Early fusion: combining RGB + thermal data (limited by AMG8833 resolution).
- Late fusion: separate detectors (camera, thermal, ultrasonic) feed a risk scoring system.
- Filtering methods: Kalman filters, Mahalanobis gating, and hysteresis thresholds reduce false positives.
- For this project: late fusion is best vision handles mid/far range, thermal aids low visibility, and ultrasonic validates near-field hazards.

Prototyping and Closed-Loop Control

- Smart Car Chassis kit: used as a train simulator with encoders and motor drivers to replicate train kinematics.
- Kinematic equations: model velocity, acceleration, and braking distance.
- Decision logic: if the fused risk score $\geq$ threshold and time-to-collision (TTC) $\leq$ safety margin, trigger escalating alerts, controlled stop.
- Operations: if the track is clear, the system computes estimated time of arrival (ETA) to the next station.

RESEARCH GAP

Despite encouraging progress, several gaps remain particularly relevant to the Sri Lankan context targeted by this project:

- Night/fog robustness with low-cost hardware: Most railway perception studies use LiDAR, RADAR, or cooled thermal cameras. Very little research explores combining low-cost RGB cameras with an AMG8833-class thermal array to achieve detection that is night-capable, powerefficient, and affordable for developing regions [1], [2], [4].
- Ultrasonic ranging on moving platforms (bias-aware integration): Existing literature and hobbyist applications usually assume static sensors. There is a gap in quantifying and compensating timeof-flight bias on moving vehicles, validating corrections with encoder-based odometry, and establishing conditions where ultrasonic should serve only as confirmatory near-field sensing rather than the main range detector [5].
- End-to-end risk scoring and braking advisories: Many studies focus solely on detection. There is limited open work that propagates perception uncertainty into a risk score, converts it into braking distance or time-to-collision (TTC) constraints, and outputs driver alerts together with operational actions (reduced speed, emergency stop) and recalculated estimated time of arrival (ETA) to stations [3].
- Novel Multimodal Fusion (low-cost stack): Combines camera-based object detection, AMG8833 thermal presence sensing, and bias-compensated HC-SR04 ultrasonic confirmation in a late fusion risk model tailored to night/fog conditions and elephant encounters.
- Dataset & Protocols: Builds a video + sensor-log dataset (RGB/thermal/ultrasonic/odometry) collected on real tracks, extending existing elephant datasets by including low-visibility cases and annotated distance labels.
- Decision Support & ETA Module: Introduces a physics-based braking and ETA recomputation engine, translating detections into actionable advisories (velocity/acceleration profiles) and updated station-to-station schedules.
- Prototype-to-Field Path: Validates the system first on a 4-wheel encoded chassis (closed-loop braking and TTC testing), then through pilot field trials (static targets near track boundaries in controlled settings), creating a reproducible baseline for Sri Lankan Railways.

RESEARCH PROBLEM

Railway transportation in Sri Lanka continues to face frequent accidents at unmanned crossings and poorly visible track segments, especially at night or during foggy conditions. Such incidents often involve pedestrians, vehicles, and wild animals (particularly elephants) that unexpectedly obstruct trains. Existing safety measures including manually operated gates, warning signs, and occasional surveillance cameras are inadequate due to their limited coverage, high deployment cost, and inability to function reliably in remote or low-visibility environments.

Although vision-based object detection using deep learning has shown promise in road traffic and urban railway contexts, its performance degrades under poor lighting and adverse weather. Thermal imaging and ultrasonic sensors can detect obstacles regardless of lighting, but each comes with limitations: low resolution in affordable thermal sensors, and distance measurement errors on moving platforms for ultrasonic modules. Moreover, most existing studies stop at detection and do not integrate multimodal sensing into a unified system that translates obstacle detection into safety actions such as braking distance computation, driver alerts, or arrival time updates.

The research problem, therefore, is the lack of a cost-effective, real-time, multimodal obstacle detection and collision avoidance system for Sri Lankan railways that can:

- Reliably detect and classify humans, vehicles, and animals under varied visibility conditions.
- Accurately estimate obstacle distance relative to a moving train.
- Integrate detection into a decision support module that computes safe braking responses and updates operational timelines.

OBJECTIVES

Main Objective

To design a real-time multimodal obstacle detection and collision avoidance system for Sri Lanka's railway network by integrating computer vision, thermal sensing, and ultrasonic ranging, thereby improving safety at unmanned crossings and low-visibility track segments.

Specific Objectives

Dataset Development

- Collect and preprocess video datasets.
- Annotate obstacles (human, elephant, vehicle, other) with bounding boxes for model training.
- Create a synchronized dataset merging video, thermal logs, ultrasonic readings, and encoder data for multimodal learning.

Model Development & Training

- Train and fine-tune YOLO-based deep learning models for railway obstacle detection.

Sensor Integration & Fusion

- Incorporate the AMG8833 infrared array for warm-body detection under fog/night.
- Integrate HC-SR04 ultrasonic sensing with encoder-based bias compensation for moving platforms.
- Develop a late fusion framework combining RGB, thermal, and ultrasonic streams into a unified risk scoring system.

Decision-Making & Control Logic

- Implement algorithms to compute braking distance, TTC (time-to-collision), and stopping margins.
- Develop a decision-support system that provides driver alerts or stop commands.
- Add an ETA recomputation module to adjust train scheduling under hazard conditions.

Prototype Development & Testing

- Build a scaled prototype on a 4-Wheel Smart Car Chassis with encoders and motor drivers.
- Validate detection accuracy, latency, braking, and reliability in varied visibility conditions.

Performance Evaluation & Validation

- Define metrics: MAP, FPS, false positives/negatives, braking success rate.
- Compare unimodal vs multimodal detection.
- Assess feasibility for cost, scalability, and robustness in Sri Lanka's railway environment.

METHODOLOGY

Overa251l System Architecture

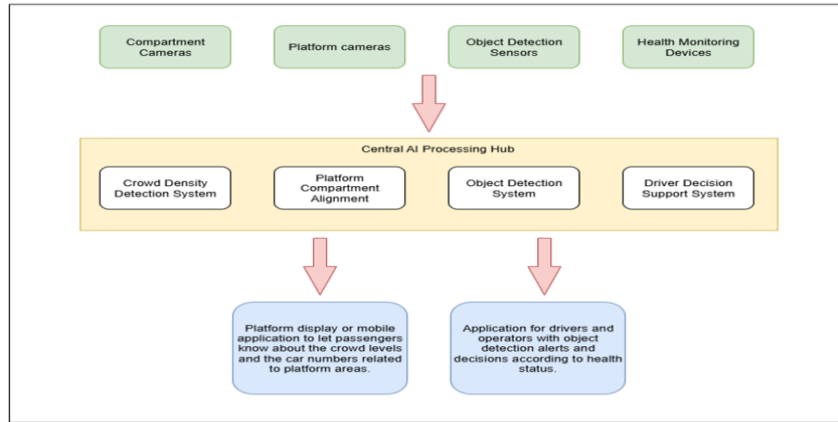


Figure 1: System Architecture Diagram

The proposed system architecture consists of four interconnected AI powered components designed to enhance railway safety and passenger management. The Compartment Crowd Density Detection module uses cameras to monitor passenger occupancy within train compartments in real time. This data directly informs the Platform Compartment Alignment module, which integrates compartment level occupancy with platform side crowd analysis to dynamically determine optimal stopping positions, ensuring efficiency and safe boarding. In parallel, the Object Detection System employs sensors to identify potential hazards and transmitting alerts to both the driver and the control center. The Driver Decision Support System integrates inputs from all modules, to calculate contextual risk scores and provide drivers with real time recommendations. In summary, these components form a unified architecture that improves passenger flow, minimizes boarding hazards, prevents collisions, and supports safer decision-making in railway operations.

Object detection system

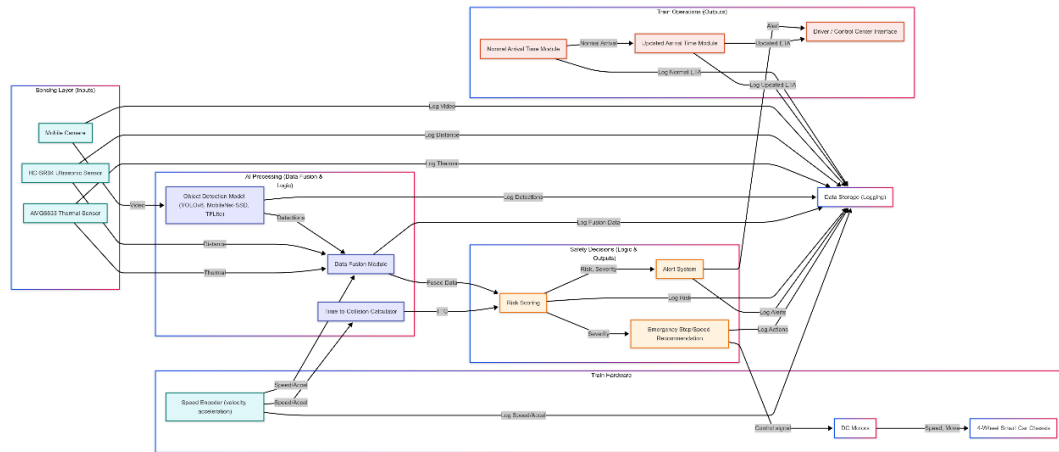


Figure 2: Object detection system Component - High Level System Architecture Diagram

The proposed solution is a multi-sensor on-train obstacle detection and alert system that enhances railway safety in Sri Lanka by identifying humans, vehicles, and wild animals (particularly elephants). It fuses computer vision, ultrasonic ranging, and thermal sensing for reliable operation in diverse scenarios: day/night, fog, and rural crossings.

Visual Object Detection Subsystem

- Camera captures continuous video of the track.
- YOLOv8-based detector identifies obstacles (humans, vehicles, elephants).
- Training uses a hybrid dataset (Wild Elephant YOLO dataset + custom recordings).
- Transfer learning from COCO weights ensures robust recognition.
- Output: bounding boxes, labels, confidence scores per frame.

Object detection distance measurement Subsystem (Ultrasonic Sensor)

- The HC-SR04 ultrasonic sensor provides near-field distance
- On a moving train, naive time-of-flight introduces errors because the vehicle advances during echo return. To mitigate:
 - Short-range measurements.
 - Speed encoder data from the prototype chassis compensates for motion bias.
- Output: real-time obstacle distance, validated against visual detections. How HC-SR04 Works for Object Detection.

The HC-SR04 is an ultrasonic distance sensor with a transmitter, receiver, and control circuit in one compact module. It works using sonar

Trigger Pulse

- The microcontroller sends a 10 μs HIGH pulse to the Trigger pin.
- This causes the sensor to emit a 40 kHz ultrasonic sound wave from its transmitter.

Wave Reflection (Echo)

- The ultrasonic wave travels through the air until it hits an object and reflects back.
- The Receiver detects this echo.

Echo Pulse Duration

- The sensor sets the Echo pin HIGH for the duration of the round trip of the sound wave.
- The microcontroller reads the time the Echo pin stays HIGH.
- Problem with Moving Train

On a moving train, the measured distance from the HC-SR04 alone is not fully accurate because:

- The train moves forward while the ultrasonic wave is traveling.
- The echo delay means the object is closer or farther than the static reading depending on train speed.

So Correct Distance Calculation While Train is Moving

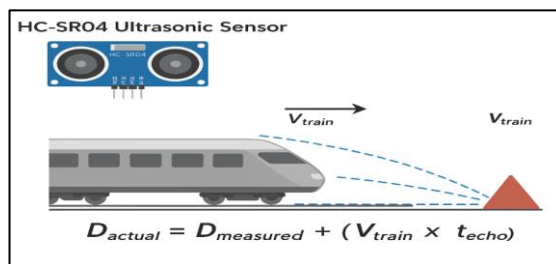


Figure 3: Train movement with sensor wave

Correct distance calculation while moving train:-

$$D_{actual} = D_{measured} + \frac{V_{train} \times t(\text{ultrasonic sensor round trip})}{2}$$

$$D_{measured} = \frac{t_s \times \text{speed of sound}}{2}$$

Figure 4: Distance calculation formula

Thermal Sensing Subsystem

- The AMG8833 infrared array (8×8 pixels) measures emitted heat from warm-bodied obstacles.
- Thermal frames are interpolated (e.g., 8×8 → 64×64 pixels) and filtered for stability. Features such as maximum pixel, centroid drift, and hot-cluster size are extracted.
- A lightweight CNN or rule-based classifier identifies human/animal/no-object presence.
- Output: binary or coarse classification with spatial cues.

Sensor Fusion and Risk Assessment

- A late-fusion architecture integrates outputs from the vision, ultrasonic, and thermal subsystems.
- Each detection is assigned a risk score based on:

Object type (e.g., elephant, human, vehicle in severity),

Distance from train

Location on/off the track

Time-to-collision (TTC) derived from velocity and range.

- If risk score $\geq$ threshold = hazard alert
- Braking & Alert Subsystem
- Alerts driver/control center when hazard detected.
- Computes safe stopping distance.
- If obstacle is within braking distance, stop command issued.
- ETA to next station updated dynamically (normal vs hazard-adjusted).

Prototype Implementation (Smart Car Simulator)

- Built on a 4-Wheel Smart Car Chassis Kit with:
- DC motors + L293D motor driver,
- 20-line encoders,

- Camera, ultrasonic, and thermal sensors at the front.
- Simulates train motion, obstacle encounters, braking, and continuation.
- Logs detections, distances, ETAs for evaluation.

PROJECT REQUIREMENTS

Functional Requirements

- Object Detection (Camera + AI)
- Capture video from a forward-facing camera.
- Detect and classify objects (human, vehicle, elephant, other) using YOLOv8 or MobileNet-SSD.
- Provide bounding boxes, labels, and confidence scores in real time (≥ 15 FPS).

Thermal Detection (AMG8833 IR Sensor)

- Detect warm-bodied objects (humans/animals) up to 7 m.
- Convert 8×8 thermal array into interpolated heatmaps.
- Provide coarse classification (human/animal/no-object).

Distance Measurement (HC-SR04 Ultrasonic Sensor)

- Measure distance to detected obstacle within 2–400 cm.
- Apply motion compensation using prototype speed encoder to reduce bias.

Sensor Fusion & Risk Assessment

- Fuse vision, thermal, and ultrasonic data.
- Assign a risk score based on object type, distance, and Time-to-Collision (TTC).
- Trigger alerts if risk score exceeds threshold.

Braking & Alert Subsystem

- Calculate safe stopping distance:
- If obstacle lies within braking range, trigger stop command.
- Send alerts to driver/control center

Station ETA Calculation

- Estimate normal arrival time between stations.
- Recalculate ETA dynamically if train slows or stops due to detected obstacles.

Prototype Validation (Smart Car Train Simulator)

- Simulate train forward motion with the 4-Wheel Smart Car Chassis.
- Demonstrate obstacle detection, braking, and ETA adjustment in real-world prototype tests

Non-Functional Requirements

Performance

- Object detection latency ≤ 100 ms per frame.
- Sensor fusion cycle ≤ 200 ms.

Robustness

- Operate reliably under low light, fog, and rain.
- Minimize false positives (e.g., tree branches not mistaken for humans).

Scalability

- System design should support deployment across multiple trains.

Energy Efficiency

- Optimized for low-power embedded devices (Raspberry Pi, Jetson Nano, Android).

Maintainability

- Modular architecture (object detection, thermal, ultrasonic, fusion, braking modules independent).

Expected Test Cases

Object Detection Tests

- Detect human on track in daylight.
- Detect elephant at night with thermal + camera.
- Detect vehicle in fog conditions.
- Ensure tree/shadow is not misclassified as a human.

Distance Measurement Tests

- Measure static obstacle at 2 m accurately.
- Measure obstacle while moving at 10 km/h with encoder compensation.
- Test cloth-covered obstacle (soft surface) → reduced accuracy flagged.

Sensor Fusion Tests

- Camera detects object but thermal does not → low risk score.
- Thermal detects object but camera misses → still trigger warning.
- Camera + ultrasonic confirm near-field obstacle → stop immediately.

Braking & Safety Tests

- Train speed = 20 km/h, obstacle at 50 m → must stop safely.
- Train speed = 40 km/h, obstacle at 30 m → must alert driver (unsafe to stop fully).
- No obstacle → train maintains speed, ETA matches expected arrival.

Prototype Validation Tests (Smart Car Train)

- Place obstacle at 10 cm → car stops before collision.
- Place obstacle at 50 cm → car slows and stops smoothly.
- Remove obstacle → car resumes and updates ETA.

BUDGET AND BUDGET JUSTIFICATION

Item	Unit Price (Rs.)
AMG8833 IR 8×8 Thermal Imager Array (MD0391)	6,850.00
HC-SR04 Ultrasonic Sensor Module (MD0026)	240.00
4-Wheel Smart Car Chassis Kit (RB0002)	1,750.00
Arduino Board	2,100.00
Total Estimated Cost	10,940.00

REFERENCES

- [1] M. Brenner, N. H. Reyes, T. Susnjak, and A. L. C. Barczak, “RGB-D And Thermal Sensor Fusion: A Systematic Literature Review,” 11 Jul 2023
[\[2305.11427\] RGB-D And Thermal Sensor Fusion: A Systematic Literature Review](#)
- [2] Wassim El Ahmar, Yahya Massoud, Dhanvin Kolhatkar, Hamzah AlGhamdi , Mohammad Alja’afreh , “Enhanced Thermal-RGB Fusion for Robust Object Detection,” CVPR Workshops, 2023..
[Enhanced Thermal-RGB Fusion for Robust Object Detection](#)
- [3] “Proceedings of the 33rd European Safety and Reliability Conference (ESREL 2023): Calculation of TTC in virtual coupling,” ESREL 2023, 2023.
<https://rpsonline.com.sg/proceedings/esrel2023/pdf/P455.pdf>
- [4] A. Marvellous, D. Saunders, M. Wannewitz, J. Fayyad, M. Pavanini, and G. Pezzano, “Cross-Modal Fusion of RGB and Thermal Data for Robust Semantic Segmentation at Night in Autonomous Vehicles,” ResearchGate, Jun. 2024.
<https://arxiv.org/pdf/2305.11427>
- [5] E. Bertolazzi, T. Fiorini, A. L. Scirocco, and M. Savarino, “Mobile robot localization via fusion of ultrasonic, inertial and odometric data,” in Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 1998, pp. 1231–1236
https://www.cs.cmu.edu/%7Emotionplanning/papers/sbp_papers/kalman/dead_kalman.pdf